

William Rudisill, Ph.D.

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11633 Snowpeak Way #504, Truckee, CA 96161

EDUCATION

Boise State University, Ph.D. Geophysics	2022
Boise State University, M.S. Hydrological Sciences	2018
University of North Carolina at Chapel Hill, B.S. Geology	2015

APPOINTMENTS

Postdoctoral Research Fellow, Lawrence Berkeley National Laboratory	2022–present
Research Scientist, ATA Aerospace	2022
Graduate Research Assistant, Boise State University	2016–2022

PROFESSIONAL ACTIVITIES

- Member, American Geophysical Union
- Manuscript Reviewer: *Journal of Hydrometeorology*, *Earth Interactions*, *Climate Dynamics*, *JGR: Atmospheres*
- NASA Terrestrial Hydrology Program Reviewer (2024)
- Co-editor, *Mountain Views Chronicle* (2024)

GRANTS AND FUNDING

Funding Agency	Title	Year	Role	Amount
US DOE Atmospheric Systems Research	<i>Surface energy balance closure studies in the Upper Colorado River Basin...</i>	2024	Co-PI	\$2M+
LBNL Earth and Environmental Sciences Area	<i>Examining the Symmetry of Daily Temperature in Complex Terrain</i>	2023	PI	\$30k

ACADEMIC AWARDS AND FELLOWSHIPS

- US DOE SCGSR Program (2022)
Examining the Hydrologic Implications of WRF Physics Parameterization Decisions in the East River, Colorado

INVITED TALKS

- University of Waterloo SUMMA Working Group, Virtual (2025): *The Future of HoarFrost in the Upper Colorado River Basin*
- NCAR Snow Affinity Group Meeting, Virtual (2024): *Are atmospheric models too cold in the mountains?*

- AGU Fall Meeting, Session GC23E-0301, Washington, D.C. (2024): *Are atmospheric models too cold in the mountains?*

CONFERENCE ORGANIZING

- Co-convener, Session C33C, AGU Fall Meeting (2025)
- Co-convener, Session C33C, AGU Fall Meeting (2024)
- Early Career Convener, Session H44C, AGU Fall Meeting (2023)

TEACHING & MENTORSHIP

- Mentor: LBNL SULI internship program (2 mentees, 2024)
- Mentor: LBNL RENEW internship program (1 mentee, 2024)
- Teaching Assistant: Hydrologic Modeling and Remote Sensing, Boise State University (2022)

TECHNICAL SKILLS

Python (pandas, xarray, dask); Geospatial analysis (gdal, geopandas, rasterio); HPC systems; Backcountry snow travel

PUBLICATIONS

- Rudisill, W.** (2025a). "Seasonality and Albedo Dependence of Cloud Radiative Forcing in the Upper Colorado River Basin". *J. Geophys. Res.* 130.6, e2024JD042366. DOI: 10.1029/2024JD042366.
- Zhou, Y. (2025). "Atmospheric river frequency-category characteristics shape U.S. west coast runoff". *J. Geophys. Res.* 130.2, e2024JD041805. DOI: 10.1029/2024jd041805.
- Rhoades, A. M. (2024). "Anticipating how rain-on-snow events will change through the 21st century: lessons from the 1997 new year's flood event". *Clim. Dyn.*, pp. 1–23. DOI: 10.1007/s00382-024-07351-7.
- Rudisill, W.** (2024b). "Are atmospheric models too cold in the mountains? The state of science and insights from the SAIL field campaign". *Bull. Am. Meteorol. Soc.* -1.aop. DOI: 10.1175/BAMS-D-23-0082.1.
- Feldman, D. R. (2023a). "The Surface Atmosphere Integrated Field Laboratory (SAIL) campaign". *Bull. Am. Meteorol. Soc.* -1.aop. DOI: 10.1175/bams-d-22-0049.1.
- Rudisill, W.**, Flores, A., Carroll, R. (2023). "Evaluating 3 decades of precipitation in the Upper Colorado River basin from a high-resolution regional climate model". *Geosci. Model Dev.* 16.22, pp. 6531–6552. DOI: 10.5194/gmd-16-6531-2023.
- Rudisill, W.** (2023a). "Cold-Season Precipitation Sensitivity to Microphysical Parameterizations: Hydrologic Evaluations Leveraging Snow Lidar Datasets". *J. Hydrometeorol.* 1.aop. DOI: 10.1175/JHM-D-22-0217.1.
- Rudisill, W.**, Kaiser, K., Flores, A. (2022). "Evaluating long-term One-Way Atmosphere-Hydrology simulations and the impacts of Two-Way coupling in four mountain watersheds". *Hydrol. Process.* 36.5. DOI: 10.1002/hyp.14578.
- Rudisill, W.**, Flores, A., McNamara, J. (2021). "The Impact of Initial Snow Conditions on the Numerical Weather Simulation of a Northern Rockies Atmospheric River". *J. Hydrometeorol.* 22.1, pp. 155–167. DOI: 10.1175/JHM-D-20-0018.1.

PUBLICATIONS IN REVIEW

- Abolafia-Rosenzweig, R. (2025). "Snow cover plays a non-dominant role in WRF/Noah-MP simulated surface air temperature cold biases over the western U.S." *JGRAtmospheres*.
- Ankor, S, **Rudisill, W.** (2025). "Impact of cloud microphysics schemes and boundary conditions on modeled snowpack in the Central Idaho Rocky Mountains, USA". *Water Resources Research*.
- Gibson, L., Feldman, D., **Rudisill, W.** (2025). "The Dual Role of Snowfall in Impacting Surface Albedo and Spring Snowmelt in the Upper Colorado River Basin". *PNAS*.
- Jackson, R. (2025). "Surface Quantitative Precipitation Estimates (SQUIRE) of snow water equivalent from the Surface Atmospheric Integrated Laboratory". *Journal of Atmospheric and Oceanic Technology*.
- Rhoades, A, North, J, **Rudisill, W.** (2025). "Snoweater heatwaves of the western United States". *Science Advances*.
- Siirila-Woodburn, E. R. (2024). "Old- and high-elevation groundwater buffers declines in mountainous runoff efficiency". *Nature Geoscience*.

PUBLISHED DATASETS

- Rudisill, W., et al. (2022). *Dynamically Downscaled (WRF) 1km, Hourly Meteorological Conditions 1987-2020. East/Taylor Watersheds*. ESS-DIVE. doi:10.15485/1845448

OPEN SOURCE CODE

- Weather Research and Forecasting Model wrapper (Python)
- WRF-Hydro Model Calibration/wrapper (python)
- Atmospheric River Identification Code (python)
- NRCS Snotel Data Downloader (python)
- Py-Ncviewer (python) *beta*

SELECT CONFERENCE PRESENTATIONS

- Feldman, D. (2025a). "Field Observations, Remote Sensing, and Modeling of Hydrology and Its Change in High-Altitude Complex Terrain". *105th AMS Annual Meeting*.
- Feldman, D. (2025b). "Super-Unit Albedo at SAIL and SPLASH and Why it Matters of Measurement-Model Intercomparisons". *ARM/ASR PI Meeting*. Vol. 2025.
- Rudisill, W.** (2025b). "Seasonality and Albedo Dependence of Cloud Radiative Forcing in the Upper Colorado River Basin". *ARM/ASR PI Meeting*. Vol. 2025.
- Born, R (2024). "Quantifying atmospheric river-driven snow accumulation and ablation: A 2022-2023 case study". *AGU Fall Meeting Abstracts*.
- Feldman, D. (2024). "Observations of super-unit albedo in complex terrain: Explaining, correcting, and interpreting this phenomenon for measurement-model intercomparisons". *AGU Fall Meeting Abstracts*. Vol. 2024, C11B-06.
- Risser, M. D. (2024). "Extremes-aware gridding of daily surface air temperature in mountains". *AGU Fall Meeting Abstracts*. Vol. 2024, GC01-67.
- Rudisill, W.**, Feldman, D. (2024). "Seasonal Changes in Surface Net Cloud Radiative Effects in the Upper Colorado River Basin". *21st Conference on Mountain Meteorology*. AMS.

- Rudisill, W.** (2024a). "Are atmospheric models too cold in the mountains?" *AGU Fall Meeting Abstracts*. Vol. 2024. 301, GC23E–0301.
- Tadigiri, K. (2024). "Characterizing atmospheric rivers and their impacts on infrastructure". *AGU Fall Meeting Abstracts*. Vol. 2024. 1127, H23I–1127.
- Zhou, Y. (2024). "The role of atmospheric river sequencing and land preconditions on US west coast runoff". *AGU Fall Meeting Abstracts*. Vol. 2024, A51D–02.
- Feldman, D. (2023b). "The analysis phase of the surface atmosphere integrated field laboratory (SAIL) field campaign begins". *AGU Fall Meeting Abstracts*. Vol. 2023. 1896, H41J–1896.
- Rudisill, W.** (2023b). "Will hoar frost in the western United States be a relic of the past?" *AGU Fall Meeting Abstracts*. Vol. 2023. 1093, C11E–1093.
- Feldman, D. (2022a). "First Light from the Surface Atmosphere Integrated Field Laboratory and their Implications for Atmosphere-through-Bedrock Science". *20th Conference on Mountain Meteorology*. AMS.
- Feldman, D. (2022b). "Surveying the first year of SAIL data and formulating hypotheses for its second year". *AGU Fall Meeting Abstracts*. Vol. 2022, H25M–1258.
- Flores, A. N. (2022). "From snowflake to snowpack: Examining the consequences of cloud microphysical representations for hydrologic uncertainty". *AGU Fall Meeting Abstracts*. Vol. 2022. ui.adsabs.harvard.edu, H22F–09.
- Feldman, D. (2021). "Using SAIL campaign measurements and integrated process modeling to better understand the headwater hydrology of the upper Colorado river basin". *AGU Fall Meeting Abstracts*. Vol. 2021, H21C–02.
- Rudisill, W.**, Flores, A., Carroll, R. (2021). "Applying hydrologic and snow data inference methods to evaluate dynamically downscaled precipitation fields". *AGU Fall Meeting Abstracts*. Vol. 2021, A53E–08.
- Flores, A. N. (2020). "Delivering of water to the critical zone and knowledge to society: Synthesizing high spatiotemporal resolution model and remote sensing data in snow-dominated, mountain watersheds". *AGU Fall Meeting Abstracts*. Vol. 2020, H225–03.
- Rudisill, W.** (2019). "A comparison of dynamically downscaled and interpolated daily meteorological datasets in the East River, CO". *AGU Fall Meeting Abstracts*. Vol. 2019, GC51L–1063.
- Rudisill, W.**, Flores, A. N. (2018). "Snow-atmosphere interactions during atmospheric rivers: A central Idaho case study". *AGU Fall Meeting Abstracts*. Vol. 2018, H12B–05B.
- FitzGerald, K. A. (2017). "Scaling up: What coupled land-atmosphere models can tell us about critical zone processes". *AGU Fall Meeting Abstracts*. Vol. 2017, H41G–1534.
- Olschanowsky, C (2017). "Developing a Collection of Composable Data Translation Software Units to Improve Efficiency and Reproducibility in Ecohydrologic Modeling Workflows". *AGU Fall Meeting Abstracts*. Vol. 2017, IN32C–06.
- Rudisill, W.** (2017). "Precipitation phase partitioning during inland penetrating atmospheric river events: Role of initial land surface conditions". *AGU Fall Meeting Abstracts*. Vol. 2017, H23F–1751.